

Foundations and Introduction to Machine Learning

Course material

Universidad de Salamanca

Faculties of Sciences and Chemical Sciences

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① Foundations

- History and evolution of machine learning
- Fundamental concepts

② Supervised Learning

- Regression
- Classification

③ Unsupervised Learning

- Clustering
- Dimensionality reduction
- Anomaly detection

Section 1

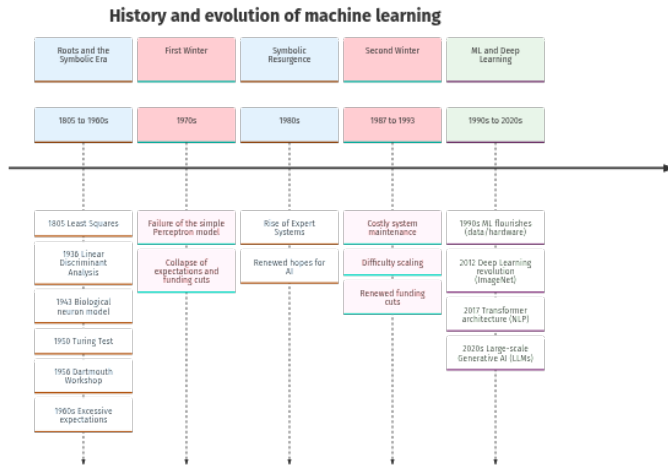
Foundations

1.1 History and evolution of ML

- Machine learning represents a **new programming paradigm**
- Instead of writing rules, the machine **infers them from data**
- Three key elements: input data, labels, and a loss function
- Its history has not been linear: cycles of enthusiasm and disillusionment

Timeline

Cycles of “summer” (breakthroughs, funding) and “winter” (disillusionment, funding cuts).



Mathematical roots and the Turing Test (1805-1950)

- 1805: Legendre and Gauss, **least squares**
- 1936: Fisher, linear discriminant analysis
- 1943: McCulloch and Pitts, first artificial neuron
- 1950: Alan Turing, "*Computing Machinery and Intelligence*" — the **Turing Test**



Source: Elliott & Fry (1951), public domain

The Perceptron and the AI winters

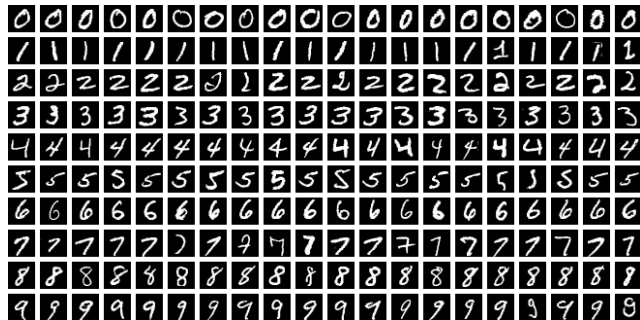
- 1958: Frank Rosenblatt creates the **Perceptron**
- **First winter** (1970s): the Perceptron cannot solve non-linear problems
- 1956: AI is formally born (Dartmouth workshop)
- **Second winter** (1990s): expert systems prove expensive and hard to scale



Source: Heinz Nixdorf MuseumsForum,

Deep Learning and the LLM era (2010-today)

- 2012: Hinton's victory in the **ImageNet** challenge
- **MNIST**: the classic benchmark for digit classification
- 2017: the **Transformer** architecture — a revolution in NLP
- 2020s: **Generative AI** (ChatGPT, Gemini) at billion-parameter scale



Source: Suvanjanprasai (2024), Wikimedia Commons, CC BY-SA 4.0

Data, features, and labels

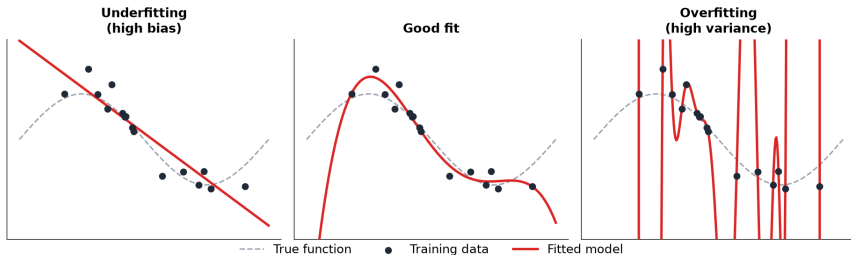
Sample	Feature 1	Feature 2	Feature 3	Label
1	25	70	175	Cat
2	8	4	90	Cat
3	35	80	180	Dog

- **Sample:** an individual data point
- **Features:** attributes describing the sample
- **Label:** what we want to predict — the “ground truth”

Overfitting vs. underfitting

- **Underfitting:** the model is too simple, misses the pattern (high bias)
- **Overfitting:** the model memorizes training noise (high variance)
- Goal: the bias-variance trade-off that generalizes best

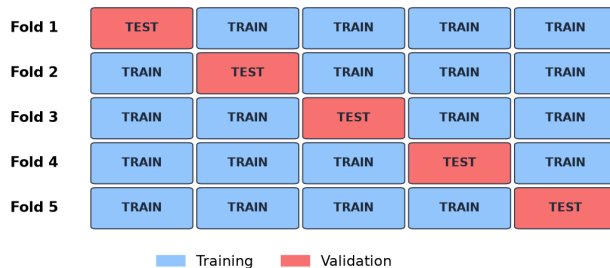
Underfitting vs. good fit vs. overfitting



K-fold cross-validation

- Splits data into K partitions (*folds*); each one acts once as the test set
- Final metric = average of the K results → a more robust estimate
- Avoids relying on a single train/test split

5-fold cross-validation



Confusion matrix and metrics

Confusion matrix

		Prediction	
		Positive	Negative
Actual class	Positive	TP True Positive	FN False Negative
	Negative	FP False Positive	TN True Negative

$$\text{Precision} = \frac{TP}{TP + FP}$$

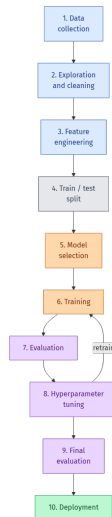
$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{P \times R}{P + R}$$

TP/TN: correct predictions; FP/FN: errors. F1 balances precision and recall, useful with imbalanced classes.

The ML *pipeline*: 10 stages

- An **iterative** process: evaluation can send you back to earlier stages
- Much of the real effort lies in stages 2-3 (data) and 7-8 (tuning)
- Deployment isn't the end: it requires continuous monitoring



Section 2

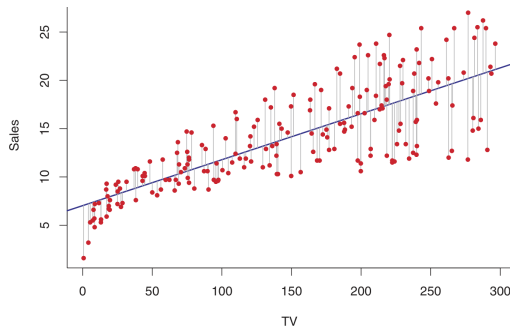
Supervised Learning

Linear regression

- Predicts a **continuous value** Y from predictors X
- Simple: $Y \approx \beta_0 + \beta_1 X$
- Multiple: $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$
- **OLS**: minimizes the residual sum of squares (RSS)

Real example: the *Advertising* dataset

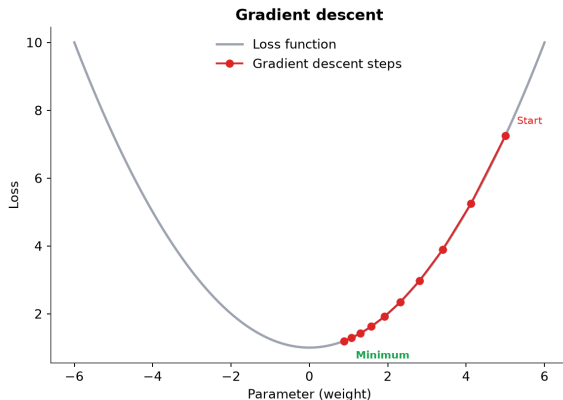
- Predicts *Sales* from TV, radio, and newspaper advertising spend
- The line minimizes the sum of squared residuals (OLS)
- Classic example of simple linear regression



Source: James, Witten, Hastie & Tibshirani (2013), *An Introduction to Statistical Learning*, Fig. 3.1. Freely distributed textbook (statlearning.com).

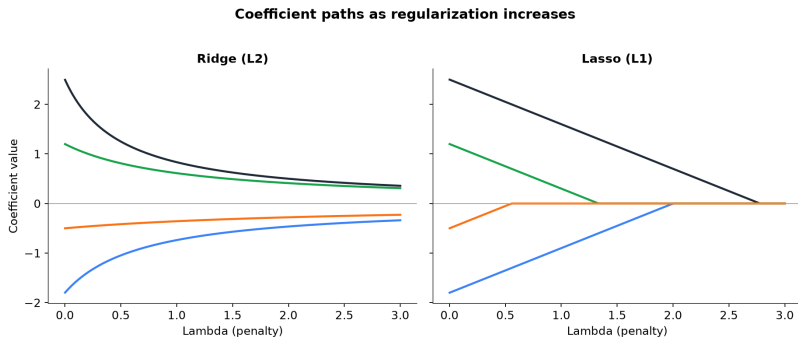
Gradient descent

- Iterative algorithm: updates parameters opposite to the gradient direction
- $\theta \leftarrow \theta - \alpha \nabla J(\theta)$, where α is the learning rate
- Too high a rate diverges; too low converges slowly



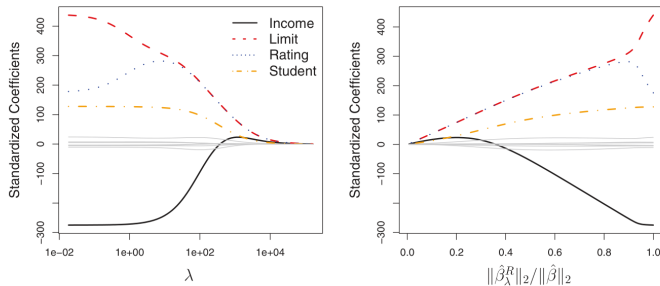
Regularization: Ridge and Lasso

- **Ridge** (L_2): penalizes $\sum \beta_j^2$, shrinks coefficients without zeroing them
- **Lasso** (L_1): penalizes $\sum |\beta_j|$, can zero out coefficients (variable selection)
- λ controls the strength of the penalty



Ridge on real data: the *Credit* dataset

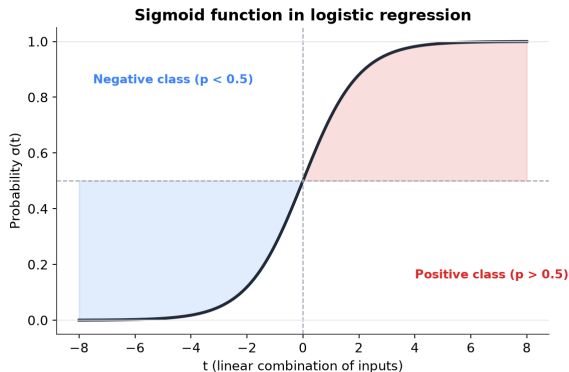
- As λ increases, coefficients shrink toward zero
- Trades a small increase in bias for a reduction in variance



Source: James, Witten, Hastie & Tibshirani (2013), *An Introduction to Statistical Learning*, Fig. 6.4.

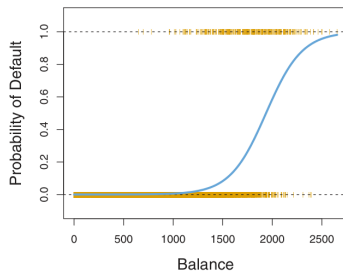
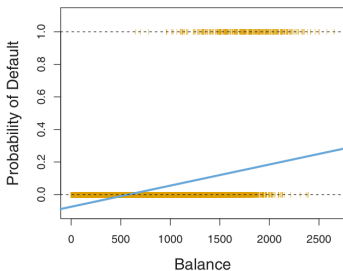
The sigmoid function

- Maps any real value to a probability between 0 and 1
- $\sigma(z) = \frac{1}{1 + e^{-z}}$
- The basis of logistic regression



Why not use linear regression?

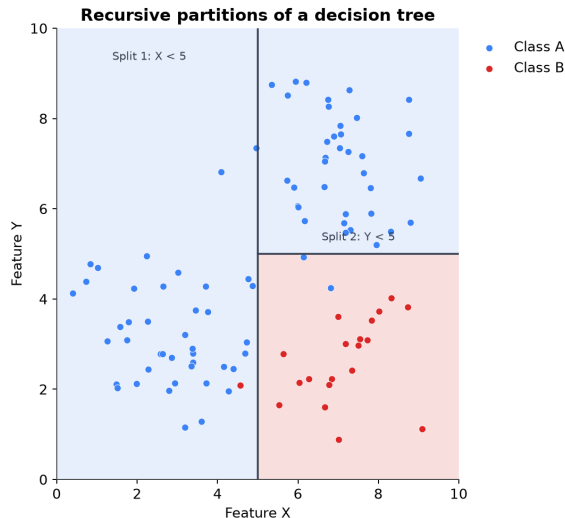
- Linear regression can predict probabilities < 0 or > 1 (nonsensical)
- Logistic regression uses the sigmoid to keep output within $[0, 1]$



Source: James, Witten, Hastie & Tibshirani (2013), *An Introduction to Statistical Learning*, Fig. 4.2.

Decision trees

- Splits feature space with successive orthogonal cuts
- Each split minimizes child-node impurity (Gini or entropy)
- Easy to interpret, but prone to overfitting without pruning

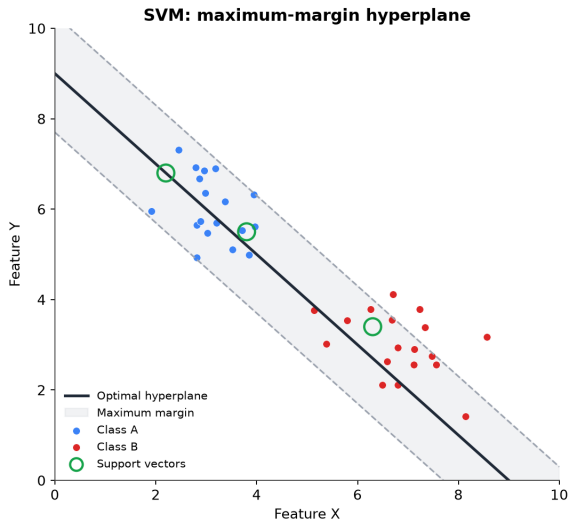


Ensembles: Random Forest and Boosting

- **Random Forest:** *bagging* + random feature subset at each split
- **Boosting:** sequential models that correct the previous one's errors (XGBoost, LightGBM)
- **Voting / Stacking:** combine the outputs of several independent models

SVM: maximum margin

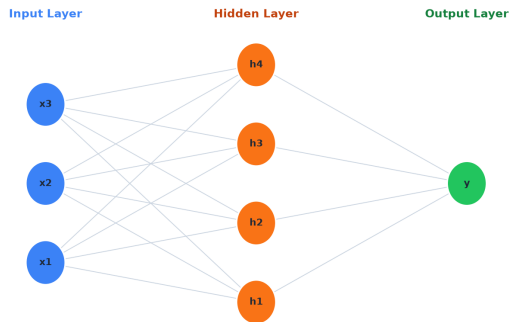
- Finds the hyperplane that maximizes the margin between classes
- Support vectors are the points closest to the hyperplane
- The *kernel trick* enables non-linear boundaries



Neural networks

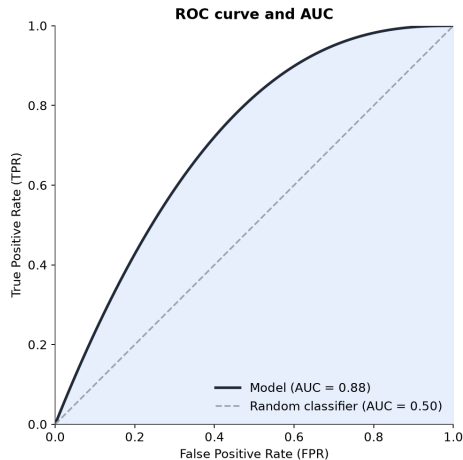
- Layers of neurons connected by adjustable weights
- Each neuron applies a non-linear activation function (ReLU, sigmoid...)
- Trained via backpropagation of the error

Architecture of a fully-connected neural network



Evaluation: ROC curve and AUC

- Y-axis: true positive rate; X-axis: false positive rate
- $AUC = 1$: perfect classifier; $AUC = 0.5$: equivalent to random guessing
- Compares models independently of the chosen threshold



Section 3

Unsupervised Learning

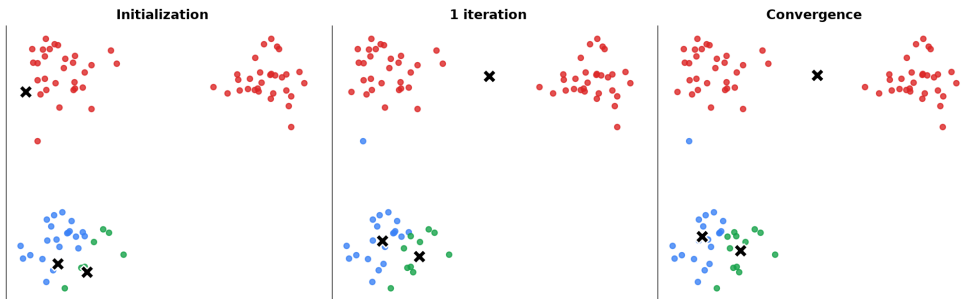
What is clustering?

- No labels: we only have X , no y
- Goal: group data with **high internal similarity** and **high external differentiation**
- Algorithms: K -means, hierarchical, DBSCAN

The iterative K -means process

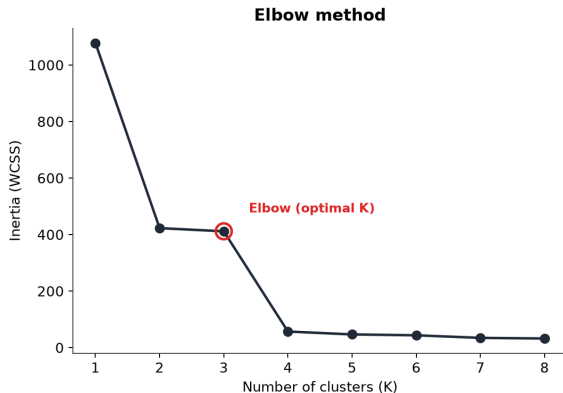
- 1 Initialize K centroids randomly
- 2 Assign each point to the nearest centroid
- 3 Recompute centroids as the mean of their cluster; repeat until convergence

Iterative K-means process



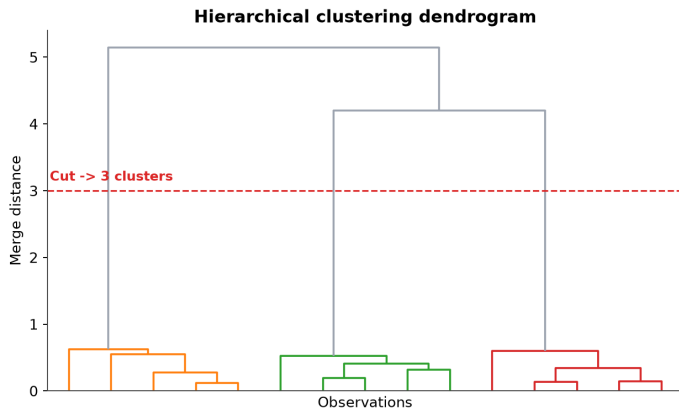
The elbow method

- Plots inertia (sum of squared distances) against K
- The “elbow” marks the point of diminishing returns: the optimal K



Hierarchical clustering: dendrogram

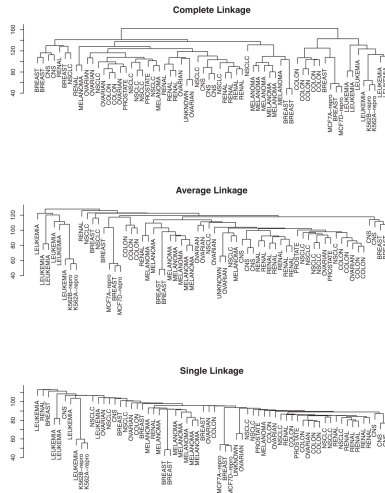
- Builds a hierarchy of clusters without fixing K in advance
- Cutting the dendrogram at a given height defines the number of clusters



Real data: NCI60 (cancer cell lines)

- Groups cancer cell lines by gene-expression similarity
- Lines of the same cancer type tend to cluster on the same branch

Source: James, Witten, Hastie & Tibshirani (2013), *An Introduction to Statistical Learning*, Fig. 10.17.

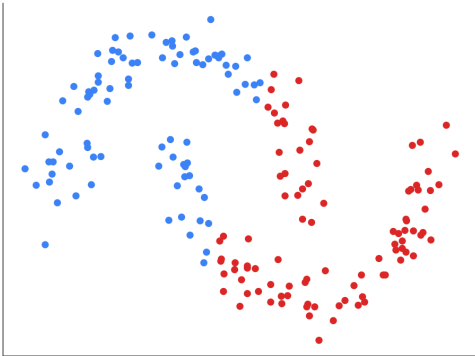


DBSCAN vs. K -means

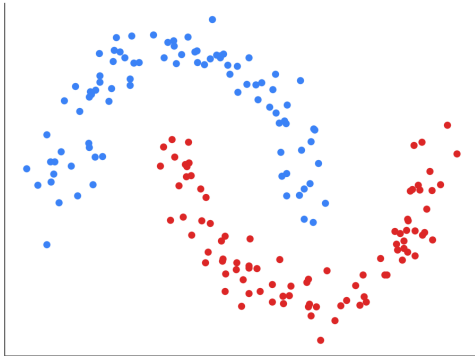
DBSCAN clusters by density: no need to fix K , it detects arbitrary shapes and automatically identifies noise (points belonging to no cluster).

K-means vs. DBSCAN on moon-shaped data

K-means (fails on non-convex shapes)

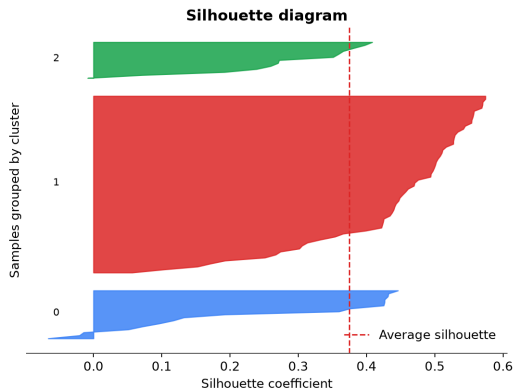


DBSCAN (detects arbitrary shapes)



Evaluation: silhouette coefficient

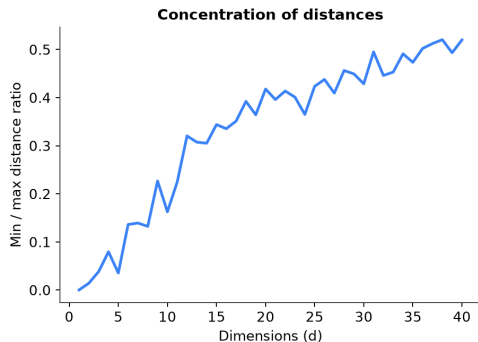
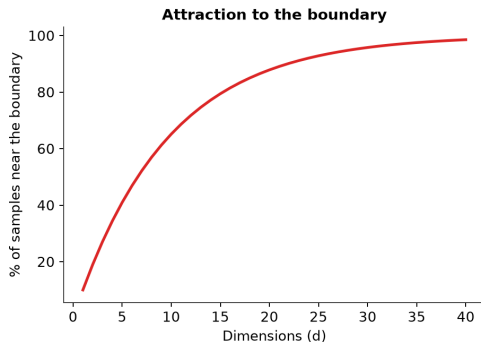
- Measures how similar a point is to its own cluster versus others
- Range $[-1, 1]$: high values indicate well-separated clusters



The curse of dimensionality

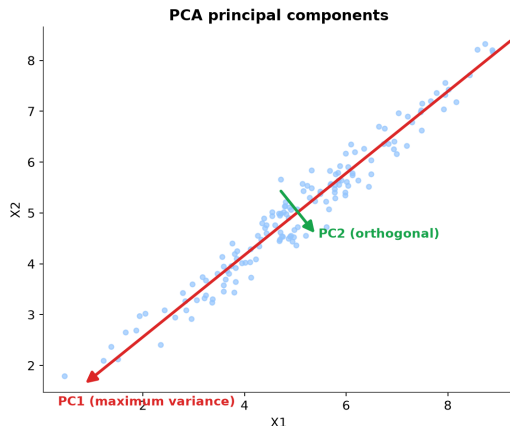
As dimensions increase, data becomes extremely sparse and distances between points become nearly indistinguishable.

The curse of dimensionality, quantified



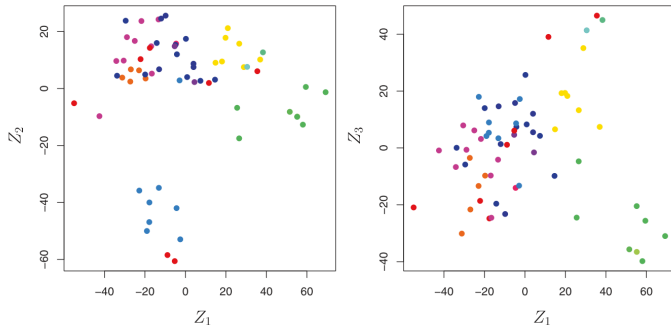
PCA: principal components

- PC1: direction of maximum variance; PC2: orthogonal, captures remaining variance
- Reduces dimensionality while preserving most of the information



PCA on real data: NCI60 (6830 genes)

From 6830 genes down to just 3 principal components: lines of the same cancer type (same color) cluster together despite the huge original dimensionality.



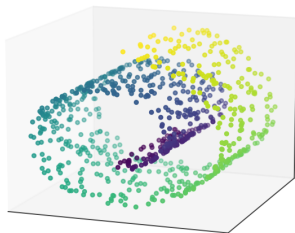
Source: James, Witten, Hastie & Tibshirani (2013), *An Introduction to Statistical Learning*, Fig. 10.15.

Manifold learning: the Swiss roll

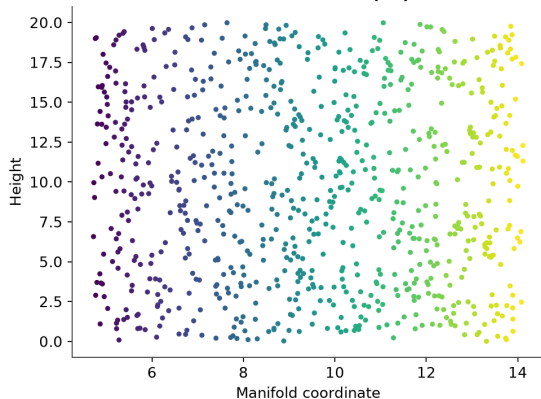
Linear PCA would collapse distant points on the rolled-up surface; *manifold learning* “unrolls” the surface while preserving local distances.

Swiss roll: the motivation for manifold learning

Original data (3D)



Unrolled manifold (2D)



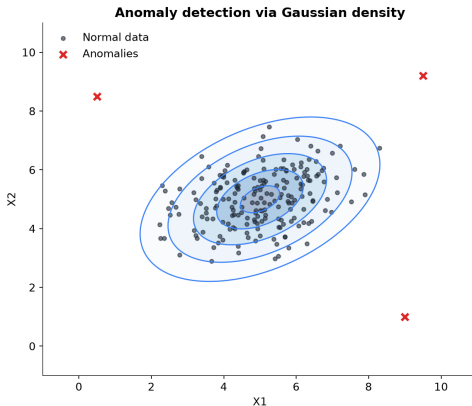
- **t-SNE**: preserves local neighborhoods; excellent for visualization, slow on large datasets
- **UMAP**: faster, better preserves global structure
- Both are non-linear — they capture relationships PCA cannot

Philosophy of anomaly detection

- *Inliers* (normal) vs. *outliers* (anomalies)
- **Anomaly detection**: contaminated training dataset
- **Novelty detection**: clean training dataset

Gaussian approach

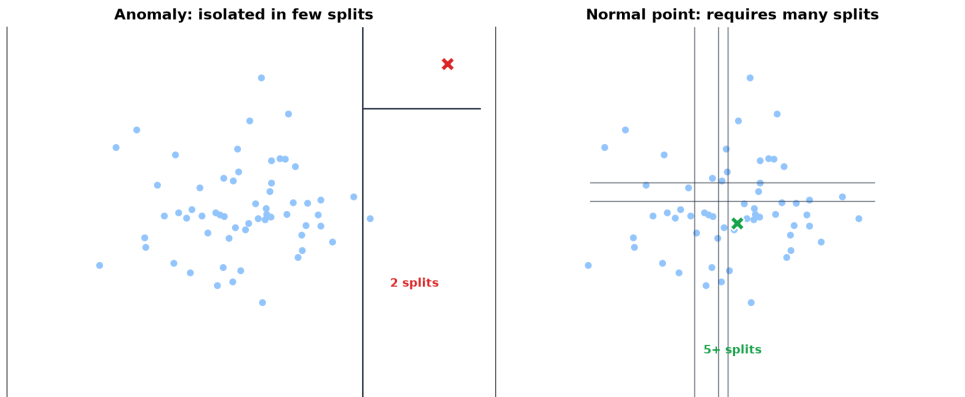
- Fits a (multivariate) Gaussian distribution to the normal data
- Points with low probability density are flagged as anomalies



Isolation Forest

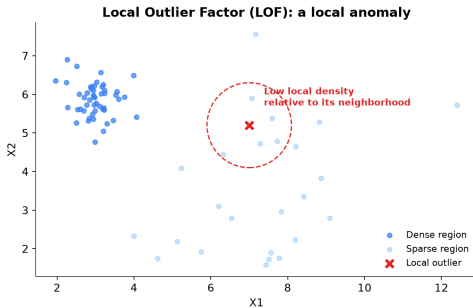
Isolates points via random partitions of the feature space: anomalies require very few partitions to become isolated.

Isolation Forest: the isolation intuition



Local Outlier Factor (LOF)

- Compares a point's local density with that of its neighbors
- Detects local anomalies that a global density threshold would miss



Foundations

- History of ML
- Overfitting/underfitting
- Cross-validation

Supervised

- Regression, regularization
- Classification, SVM
- Neural networks

Unsupervised

- Clustering
- PCA, manifold learning
- Anomaly detection

Thank you!

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